

Triangles, radians and sectors - Set 1

TMUA practice questions by topic. Specification MM4.1, MM4.2, M5.16. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 In triangle ABC , $AB = 5$, $AC = 8$ and angle $BAC = 60^\circ$. Find BC .

A $\sqrt{129}$
B $\sqrt{49 + 40\sqrt{3}}$
C $\sqrt{89}$
D $\sqrt{69}$
E 9
F 7

- 2 A triangle has sides 4, 5 and 6. Find the cosine of its largest angle.

A $\frac{9}{16}$
B $\frac{8}{9}$
C $8\frac{1}{8}$
D $-\frac{1}{8}$
E $-\frac{4}{3}$
F $-\frac{4}{4}$

- 3 A sector of a circle of radius 6 cm has area $15\pi \text{ cm}^2$. Find the arc length of the sector.

A $\frac{5\pi}{6} \text{ cm}$
B $10\pi \text{ cm}$
C $5\pi \text{ cm}$
D $\frac{5\pi}{2} \text{ cm}$
E $3\pi \text{ cm}$
F $15\pi \text{ cm}$

4 Convert 135° to radians.

- A $\frac{4\pi}{3}$
- B $\frac{3\pi}{4}$
- C $\frac{8\pi}{5}$
- D $\frac{4\pi}{5}$
- E $\frac{2\pi}{3}$
- F $\frac{3\pi}{4}$

5 Two sides of a triangle are 3 and 5 and the included angle is θ , where $\cos \theta = -\frac{1}{5}$. Find the length of the third side.

- A $\sqrt{34}$
- B 6
- C $\sqrt{40}$
- D 7
- E $\sqrt{28}$
- F $\sqrt{31}$

6 The minute hand of a clock is 12 cm long. How far does its tip travel in 25 minutes?

- A 25π cm
- B 5π cm
- C 10π cm
- D 12π cm
- E $\frac{25\pi}{6}$ cm
- F 20π cm

7 In triangle ABC , P : ' $a^2 = b^2 + c^2$ '. Q : 'angle $A = 90^\circ$ '. Which one of the following is true?

- A P is necessary and sufficient for Q.
- B P is necessary but not sufficient for Q.
- C P is sufficient but not necessary for Q.
- D P is neither necessary nor sufficient for Q.

- 8 Triangle PQR has area $10\sqrt{3}$, $PQ = 8$ and $PR = 5$. Given that angle QPR is obtuse, find QR .
- A 13
 B $\sqrt{89}$
 C $\sqrt{89 + 40\sqrt{3}}$
 D $\sqrt{129}$
 E 7
 F $\sqrt{109}$
- 9 A chord of a circle of radius 4 subtends an angle of $\frac{\pi}{3}$ at the centre. Find the area of the minor segment cut off by the chord.
- A $\frac{8\pi}{3} - 8\sqrt{3}$
 B $\frac{8\pi}{3} - 4$
 C $\frac{16\pi}{3} - 4\sqrt{3}$
 D $8\pi - 4\sqrt{3}$
 E $\frac{8\pi}{3} - 4\sqrt{3}$
 F $\frac{4\pi}{3} - 2\sqrt{3}$
- 10 In triangle ABC , $AB = 6$, $BC = 4$ and angle $BAC = 30^\circ$. How many possible triangles are there?
- A 1
 B 3
 C 2
 D 0
 E infinitely many
- 11 The perimeter of a sector of radius r and angle θ radians is 20, and its area is 16. Find the possible values of r .
- A 4 only
 B 8 only
 C 1 or 16
 D 2 or 8
 E 4 or 8
 F 2 or 4

12 In triangle ABC , $\sin A : \sin B : \sin C = 3 : 5 : 7$. Find angle C .

A 120°

B 150°

C 100°

D 135°

E 90°

F 60°